

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/258040673>

Performance of Maximum Number of Repetitions With Cluster-Set Configuration

Article in *International Journal of Sports Physiology and Performance* · October 2013

DOI: 10.1123/IJSP.2013-0246 · Source: PubMed

CITATIONS

30

READS

1,148

5 authors, including:



Eliseo Iglesias-Soler

University of A Coruña

146 PUBLICATIONS 637 CITATIONS

[SEE PROFILE](#)



Edu Carballeira

University of A Coruña

61 PUBLICATIONS 273 CITATIONS

[SEE PROFILE](#)



Tania Sánchez-Otero

University of A Coruña

11 PUBLICATIONS 110 CITATIONS

[SEE PROFILE](#)



Xian Mayo

King Juan Carlos University

79 PUBLICATIONS 368 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



Physiological and motor engagement in Physical Education. Factors associated [View project](#)



Prediction dropout in Fitness centres [View project](#)

Performance of Maximum Number of Repetitions With Cluster-Set Configuration

Eliseo Iglesias-Soler, Eduardo Carballeira, Tania Sánchez-Otero,
Xian Mayo, and Miguel Fernández-del-Olmo

Purpose: To analyze performance during the execution of a maximum number of repetitions (MNR) in a cluster-set configuration. **Method:** Nine judokas performed 2 sessions of parallel squats with a load corresponding to 4-repetition maximum (4RM) with a traditional-training (TT) and cluster-training (CT) set configuration. The TT consisted of 3 sets of repetitions leading to failure and 3 min of rest between sets. In the CT the MNR was performed with a rest interval between repetitions (45.44 ± 11.89 s). The work-to-rest ratio was similar for CT and TT. **Results:** MNR in CT was 45.5 ± 32 repetitions and was 9.33 ± 1.87 times the volume in TT. There was a tendency for the average mean propulsive velocity (MPV) to be higher in CT (0.39 ± 0.04 vs 0.36 ± 0.04 m/s for CT and TT, respectively, $P = .054$, standardized mean difference [d] = 0.57). The average MPV was higher in CT for a similar number of repetitions (0.44 ± 0.08 vs 0.36 ± 0.04 m/s for CT and TT, respectively, $P = .006$, $d = 1.33$). The number of repetitions in TT was correlated with absolute 4RM load ($r = -.719$, $P = .031$) but not in CT ($r = -.273$, $P = .477$). **Conclusions:** A cluster-set configuration allows for a higher number of repetitions and improved sustainability of mechanical performance. CT, unlike TT, was not affected by absolute load, suggesting an improvement of training volume with high absolute loads.

Keywords: resistance training, interrepetition rest, muscle endurance, work-to-rest ratio, mechanical performance

Adaptation to resistance training is mediated by the interaction between mechanical, hormonal, and metabolic stimuli.^{1,2} Several training variables such as load, volume (sets $\times$ repetition), frequency of training, speed of contraction, work-to-rest ratios, or time under tension have a direct influence on the magnitude of these stimuli.³ These variables are interdependent, so an increment of one factor can decrease another variable. In this regard the relationship between the number of repetitions performed in each set and the maximum possible number of repetitions (MNR) is an important factor that is associated with intensity, volume, and metabolic response to a workload.^{4,5} It has been shown that traditional set configuration with repetitions leading to failure results in a decline in mechanical performance.⁶ This requires long rest periods to sustain the number of repetitions in successive sets⁵ or reductions in the training load to maintain the initial number of repetitions.^{7,8} All these acute effects decrease both the mechanical stimuli and the volume of the session (load $\times$ repetition $\times$ load).

An alternative set configuration consists of manipulating work and rest periods by breaking sets into small clusters of repetitions. This type of training has been termed cluster training (CT), interrepetition rest training, or intraset rest loading.⁹⁻¹² The theoretical basis of cluster-set configuration lies in the short rest periods between clusters of repetitions, allowing replenishment of muscle phosphocreatine, improving the quality of each effort and of subsequent training adaptation.¹¹ In this regard, previous studies have shown that CT involves higher sustainability of mechanical performance,^{10,12-14} improved technique in complex resistance exercises,¹⁵ and reduced metabolic demand.^{2,14}

A question that must be clarified regarding CT is the MNR that can be completed before failure. This issue is of interest since the volume of repetitions has been proposed to be the main variable mediating adaptation to resistance exercise.¹⁶⁻¹⁸ To our knowledge, only 1 study¹⁹ has analyzed MNR with cluster-set configuration. That work suggested that employment of 30 seconds of rest between repetitions can be used to reach a high volume (repetitions) and a high intensity (90% 1RM) during the performance of bench press and unilateral biceps curl. However, that study also showed high intersubject variability for the MNR, with coefficients of variation of 50.6% and 69.3% for bench press and biceps curl, respectively. This variability could be explained by the different individual recovery capacities and the duration of pauses between repetitions. Adjusting the pauses between repetitions to the individual recovery capacity could result in a less variable MNR during a CT protocol. Although most of the studies that have compared CT with traditional training (TT) protocols equated both load and exercise-to-rest ratio between conditions,^{12,20} rest intervals were the same for every subject. One procedure that can be used to individualize the pauses between repetitions is to use the volume completed during successive sets of repetitions until failure as a reference and to then distribute the total time of rest between repetitions.¹⁴ Hence, the work-to-rest ratio during CT is the same, but the rest interval between repetitions is adjusted for every subject according to individual muscle endurance. To our knowledge only 1 previous study individualized the rest pause in CT, addressing the number of repetitions performed in a TT.¹⁴ Those authors showed that CT allowed athletes to increase velocity compared with TT when the number of repetitions and total resting time were equated between conditions. However, the total volume that could be performed in CT was not assessed.

Therefore, the main goals of this study were to assess MNR in a cluster-set configuration when pauses are fixed in reference to

The authors are with the Dept of Physical Education and Sports, University of A Coruña, A Coruña, Spain. Address correspondence to eliseo@udc.es.

individual performance during a session of sets of repetitions to failure and to compare performances between MNR in CT and TT, trying to identify variables in TT that affect MNR in CT. Thus, we investigated the evolution of velocity along quintiles of MNR in CT (ie, repetitions that represented 20%, 40%, 60%, 80%, and 100% of total volume) compared with TT. In addition we evaluated MNR in CT with respect to the mechanical performance (ie, velocity) and analyzed the relationship between accumulated number of repetitions in successive sets of TT and MNR in CT. We hypothesized that a relationship would exist between these 2 types of muscle endurance and, therefore, individualizing rest intervals with respect to the volume completed during TT could result in better mechanical performance during CT.

Materials and Methods

Subjects

Nine trained male judoists (age 23.8 ± 4.1 y, height 176 ± 8 cm, weight 84 ± 17 kg) participated in the study. Subjects had trained with weights a minimum of 2 d/wk for at least 3 years. All the subjects volunteered and provided written informed consent. This study was approved by the local ethics committee of the University of A Coruña.

Experimental Design

In a repeated-measures design, athletes completed 4 sessions of the experimental protocol with a 72-hour interval between sessions. A parallel back-squat exercise was performed in each session. Every session started with a warm-up of 5 minutes of running on a treadmill at 50% to 70% of maximum heart rate, followed by a

5-minute session of calisthenic exercises. We evaluated the 1-repetition maximum (1RM) in the parallel squat in the first session. We employed the 1RM protocol based on the velocity decrement with increase of load; this protocol has been described previously.¹⁴ The heaviest load that each subject was able to lift only once was considered 1RM. Then we obtained the load corresponding to the 4-repetition maximum (4RM) in the second session. The 4RM load was calculated following a protocol previously described¹⁴ and was defined as the load that the participant was able to properly lift a maximum of 4 times. In the third session athletes performed a session (TT) that consisted of 3 sets leading to failure with the 4RM load and with 180 seconds of rest interspersed between sets (360 s of total recovery time). The objective of this session was to obtain the MNR that the athletes were able to complete with a traditional structure; therefore, it was not possible to randomize the sessions. Finally, during a fourth testing session, every subject performed multiple sets of 1 repetition with a 4RM load until the point of exhaustion. In this case rest between repetitions was calculated based on the ratio of the total recovery time during TT (360 s) to the individual number of repetitions completed in TT minus 1. An example of this procedure is illustrated in Figure 1.

Measurement of Dynamic Performance

Parallel back squats were performed using a standard Smith Machine (Life Fitness, Brunswick Corp, USA). Athletes executed parallel squats with a knee-joint flexion of up to 90°. We encouraged them to perform the exercise in an explosive manner and to reach the 90° knee-flexion position. Range of movement was controlled by a linear encoder (MuscleLab, Bosco Systems, Norway) attached to the bar that displayed on a monitor and a scroll bar that tracked the displacement of the load. Thus, the scroll bar had to reach in every

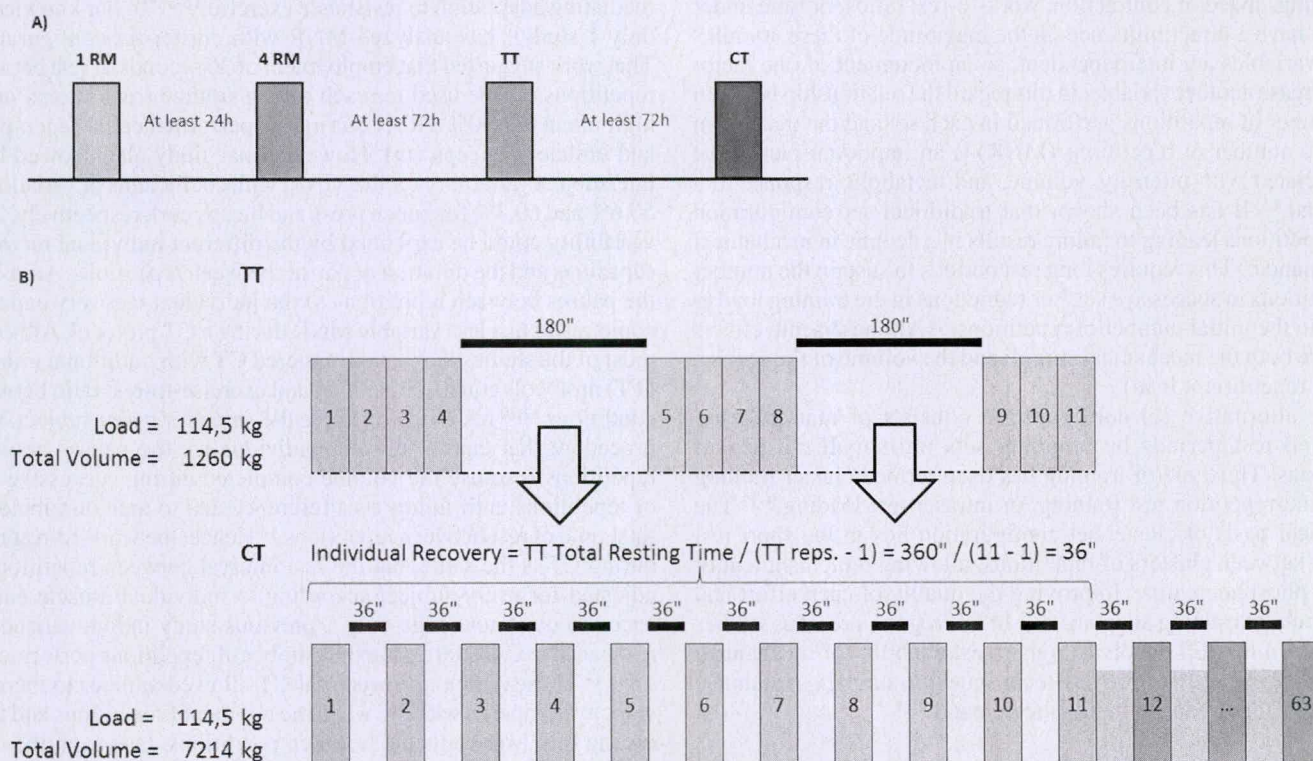


Figure 1 — (A) Schematic representation of the study and (B) example of evaluation of rest intervals in cluster training (CT). The athlete performed 11 repetitions in traditional training (TT). To accumulate 360 s of recovery at the 11th repetition of CT, we needed to distribute the total resting time between the previous 10 recovery intervals. Abbreviations: reps. indicates repetitions. Total volume = repetitions $\times$ load.

repetition the lower position showed on the screen. The position was measured with accuracy lower than 0.075 mm, and signals were sampled at a frequency of 100 Hz. Bosco et al²¹ reported for this device a measurement error of velocity due to the system lower than 0.9%, a relative reliability of .97, and an absolute reliability (coefficient of variation) of 2.3%.

Another dynamic measurement device (T-Force System, Ergotech, Murcia, Spain) was employed to evaluate velocity and concentric displacement. This system consists of a linear velocity transducer interfaced to a personal computer by means of a 14-bit-resolution analog-to-digital data-acquisition board and custom software. Instantaneous velocity is sampled at a frequency of 1000 Hz and subsequently smoothed with a fourth-order low-pass Butterworth filter with a cutoff frequency of 10 Hz. A digital filter with no phase shift was then applied to the data. Validity and reliability of this device have been previously reported.⁴ Briefly, the mean relative error in the velocity measurements was found to be lower than 0.25%, whereas displacement was accurate to ± 0.5 mm. In addition, the intraclass correlation coefficient between devices was 1.00 (95% CI 1.00–1.00) and the coefficient of variation was 0.57% for mean propulsive velocity.

During TT and CT sessions the same exercise execution criteria were kept to analyze mean propulsive velocity (MPV) of every repetition, that is, the mean velocity during the propulsive phase. The propulsive phase was defined as the portion of the concentric period during which barbell acceleration is greater than acceleration due to gravity (ie, > -9.81 m/s²).⁴ Variability of performance in every session was tested by standard deviation of MPV.

Statistical Analyses

SPSS version 15.0 software (SPSS, Inc, Chicago, IL, USA) was used to analyze the data. Descriptive parameters are shown as mean $\pm$ SD. Normal distribution of parameters was tested using the Kolmogorov-Smirnov (Lilliefors) test. To compare performance of TT and CT, a paired *t* test was used. One-way repeated-measures ANOVA of 6 levels (first repetition and those corresponding to 20%, 40%, 60%, 80%, and 100% of completed number of repetitions) was employed to compare MPV across CT. One-way repeated-measures ANOVA of 5 levels was applied to compare percentage changes of repetition corresponding to 20%, 40%, 60%, 80%, and 100% of total volume. Post hoc analyses were conducted using paired *t* tests with a Bonferroni adjustment for multiple comparisons. Effect size was calculated using standardized mean differences (*d*). 95% confidence intervals of *d* were obtained according to asymptotic estimates of standard error for paired repeated measures.²² To identify factors that could affect MNR in CT, correlations were performed using Pearson linear correlation coefficients if normality of variables could be assumed and Spearman rho if not. Significance level was set at $P \leq .05$. A post hoc power analysis was calculated using the G Power software (version 3.1.4). Statistical power for a sample size of 9 and a large effect size ($d = 0.8$) for a paired *t* test was 0.559.

Results

Number of Repetitions: TT Versus CT

Table 1 shows data corresponding to completed volume (repetitions) and volume load (repetitions $\times$ load) during TT and CT. As expected, both parameters were significantly higher in CT than in TT ($P = .008$, $d = 1.64$, and $P = .009$, $d = 1.52$, for volume load and number of repetitions, respectively). There was high variability of

Table 1 Volume Achieved During Performance of Maximum Number of Repetitions With a Cluster Set and in Traditional Configurations of Sets to Failure, Mean $\pm$ SD

Variable	Volume
1RM load (kg)	128.72 $\pm$ 19.58
1RM/BM	1.57 $\pm$ 0.34
1RM/BM ^{0.67}	6.71 $\pm$ 1.23
4RM load (kg)	112.83 $\pm$ 20.27
4RM/1RM (%)	87.56 $\pm$ 6.47
4RM/BM	1.39 $\pm$ 0.38
4RM/BM ^{0.67}	5.92 $\pm$ 1.38
Reps. CT	45.5 $\pm$ 32
Reps. TT	9.33 $\pm$ 1.87
Volume CT (kg)	5025.11 $\pm$ 3443.56
Volume TT (kg)	1028.11 $\pm$ 146.18
Reps. set 1	4.55 $\pm$ 0.88
Reps. set 2	2.67 $\pm$ 1.41
Reps. set 3	2.11 $\pm$ 1.05
Reps. CT/reps. TT	4.92 $\pm$ 3.48
Reps. CT/reps. set 1	10.78 $\pm$ 8.41

Abbreviations: 1RM indicates 1-repetition maximum; BM, body mass; 4RM, 4-repetition maximum; 1RM/BM, 1RM in ratio scaling; 1RM/BM^{0.67}, 1RM in allometric scaling; 4RM/1RM, percentage of 4RM respect to 1RM; 4RM/BM, 4RM in ratio scaling; 4RM/BM^{0.67}, 4RM in allometric scaling; Reps., repetitions; CT, cluster training; TT, traditional training of sets to failure; volume, reps. $\times$ load; Reps. CT, maximum number of reps. performed with cluster set configuration; Reps. TT, number of reps. performed in traditional training of sets to failure; Reps. set 1, number of reps. in set 1 of TT; Reps. set 2, number of reps. in set 2 of TT; Reps. set 3, number of reps. in set 3 of TT; Reps. CT/reps. TT, ratio of maximum number of reps. performed with cluster-set configuration to number of reps. completed in TT; Reps. CT/reps. set 1, ratio of number of reps. performed with cluster-set configuration to number of reps. corresponding to set 1 of TT.

MNR in CT with a coefficient of variation (CV) of 70.32%, while CV of number of repetitions performed during TT was 20.04%. In this regard, variability of the number of repetitions increased across TT (CVs of 19.34%, 52.81%, and 49.76% for sets 1, 2, and 3, respectively). The ratio between MNR in CT and number of repetitions during both set 1 and pooled data of TT are also reported in Table 1.

Mechanical Performance in CT

MPV of the first repetition and repetition corresponding to 20%, 40%, 60%, 80%, and 100% of MNR in CT are illustrated in Figure 1. One-way repeated-measures ANOVA showed a significant effect of number of repetitions ($P < .001$). Post hoc contrasts were significant between first repetition and repetitions corresponding to both 60% ($P = .008$, $d = 1.40$) and 100% ($P = .002$, $d = 1.81$) of MNR (see Figure 2). MPV was also higher in repetition corresponding to 20% of MNR with respect to the last repetition ($P = .044$, $d = 1.39$).

Percentage changes in MPV with respect to the first repetition are represented in Figure 3. Repeated-measures ANOVA showed a significant effect of percentage of MNR ($P < .001$). There was a significant difference between the last repetition (100%) and the repetition corresponding to 20% of MNR ($P = .032$, $d = 1.86$).

The effect size (d) with respect to the first repetition used to evaluate the relative volume at which the standardized mean difference was large ($d > 0.80$). A large effect was observed for repetitions corresponding to 60% to 100% of the total number of repetitions (see Figure 4).

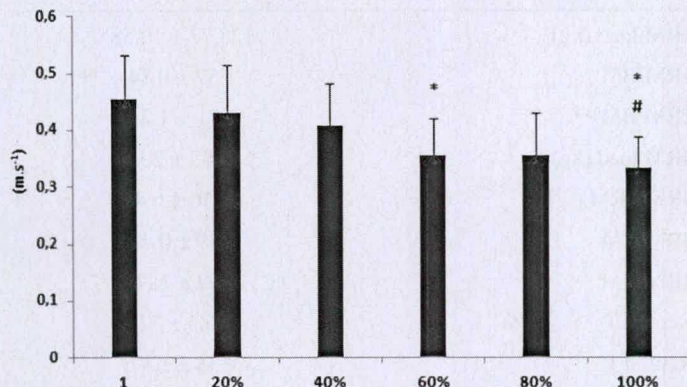


Figure 2 — Mean propulsive velocity of first repetition (1) and those corresponding to 20%, 40%, 60%, 80%, and 100% of maximum number of repetitions performed with cluster-set configuration. * $P < .01$ with respect to first repetition; # $P < .05$ with respect to 20%.

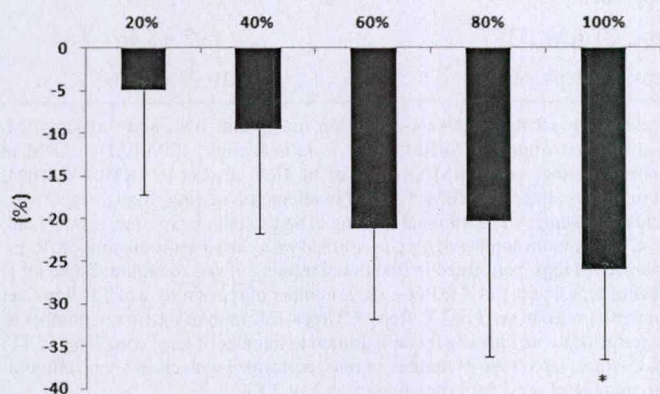


Figure 3 — Percentage change from first repetition at volumes corresponding to 20%, 40%, 60%, 80%, and 100% of maximum number of repetitions performed with cluster-set configuration. * $P < .05$ with respect to 20%.

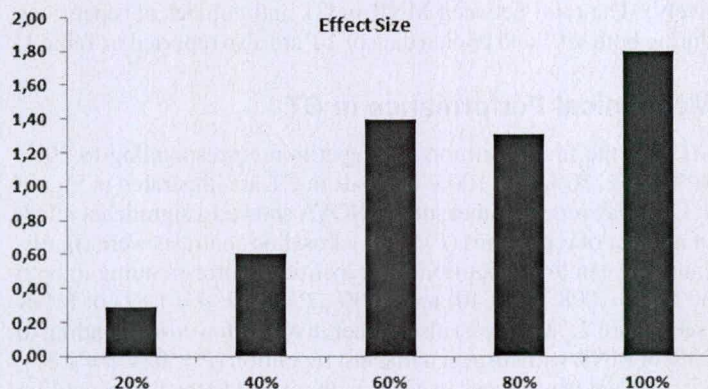


Figure 4 — Standardized mean differences with respect to the first repetition at volumes corresponding to 20%, 40%, 60%, 80%, and 100% of maximum number of repetitions performed with cluster-set configuration.

Comparison Between CT and TT Performance

Table 2 shows contrast between TT and CT with respect to mechanical performance. Although the number of repetitions was higher in CT than in TT, the average MPV was higher in CT than in TT for both the total session and for the equated performance (ie, number of repetitions corresponding to TT). Neither mean concentric displacement nor velocity decrement was significantly different between sessions.

MPVs of the last repetition performed during TT (0.30 ± 0.06 , 0.29 ± 0.08 , and 0.31 ± 0.07 m/s for sets 1, 2, and 3, respectively) and CT (0.33 ± 0.06 m/s) were not significantly different ($P = .398$). Similarly, repeated-measures ANOVA did not detect significant differences ($P = .150$) in the percentage of change in MPV between the first and last repetitions in CT and TT ($-19.12\% \pm 19.90\%$ and $-11.91\% \pm 12.35\%$ for sets 2 and 3, respectively).

Correlations Between CT and TT Performance

The number of repetitions in TT and resting time between repetitions in CT were not correlated with MNR of CT ($r = -.186$, $r = .141$, and $P = .632$, $P = .718$, for resting time and repetitions in TT, respectively). The absolute load (4RM) was negatively correlated with the number of repetitions performed in TT ($r = -.742$, $P = .022$). However, this was not the case in CT ($r = -.273$, $P = .477$). Similarly, 1RM was negatively correlated with number of repetitions in TT ($r = -.714$, $P = .031$) but not in CT ($r = -.273$, $P = .477$). In CT only, variability of MPV (standard deviation of MPV in CT) was correlated with MNR ($r = .698$, $P = .036$).

Discussion

The main findings of this study were that (1) with CT a high number of repetitions before muscular failure was completed; (2) high intersubject variability was observed with respect to the number of repetitions performed in CT; (3) performance during TT was not correlated with MNR of CT; (4) absolute load was negatively correlated with volume completed during TT but not in CT; (5) results revealed a tendency toward a higher average MPV in CT, although the volume of TT was notably lower; and (6) when only the number of repetitions corresponding to the volume performed in TT was considered, average MPV was higher in CT.

Data from our study show that the cluster-set configuration is useful to carry out high volumes of high-intensity parallel squats. This is in line with previous reports of bench presses.¹⁹ This result was expected since the last repetition before failure is performed with similar mean velocity,⁶ and CT allows a higher sustainability of velocity performance with respect to TT.^{13,14} We did not find significant differences in MPV between the last repetition of sets 1, 2, and 3 of TT and the last repetition of CT. This supports previous findings showing that mean velocity of repetitions before failure is similar for different percentages of 1RM.⁶ In the current study, percentage of change of MPV between first and last repetitions was similar for sets of TT and CT. Taking into account that the volume performed during CT was notably higher than during TT, it seems that the rate of velocity of loss was attenuated in the cluster-set configuration. Analysis of the MPV in CT showed a decrease in performance at repetitions that corresponded to 60% of MNR, and there was an increase of the standardized mean difference for repetitions corresponding to 60% of MN and above. Therefore, we can conclude that mechanical performance was reasonably maintained until 60% of total volume was achieved.

Table 2 Comparison of Performance in Cluster Training and Traditional Training, Mean $\pm$ SD

Variable	Traditional training	Cluster training	P	d (95% CI)
MPV-S (m/s)	0.36 $\pm$ 0.04	0.39 $\pm$ 0.06	.054	0.57 (0.30, 0.84)
MPV-VTT (m/s)	0.36 $\pm$ 0.04	0.44 $\pm$ 0.08	.006	1.33 (0.86, 1.19)
Concentric displacement (cm)	45.70 $\pm$ 6.25	46.86 $\pm$ 6.17	.133	0.19 (0.07, 0.31)
SD concentric displacement (cm)	2.49 $\pm$ 0.88	2.31 $\pm$ 0.39	.427	0.26 (0.02, 0.51)
$ \Delta\% $	26.12 $\pm$ 10.43	26.07 $\pm$ 11.68	.990	0.004 (-0.28, 0.29)

Abbreviations: CI indicates confidence interval; MPV, mean propulsive velocity; MPV-S, average MPV of every session; MPV-VTT, average MPV until achieving number of repetitions performed in traditional; $|\Delta\%|$, absolute value of percentage change from first to last repetition.

It has been previously suggested that improvement of mechanical performance observed with cluster-set configuration could be caused by partial replenishment of phosphocreatine (PCr) in the working muscle during rest periods between repetitions.^{2,10,12,23} Mean rest pause in the current study (45.44 $\pm$ 11.89 s) should allow at least 50% of PCr recovery²³ and therefore could result in reduced loss of velocity that usually occurs across a set of resistance exercises.⁶ High correlations between loss of velocity and metabolic measures of fatigue have been reported,⁴ supporting the idea that CT delays the metabolic stress associated with muscle failure. Moreover, despite the fact that the volume of CT was notably higher than in TT, a tendency ($P = .054$) for higher MPV in CT was observed. Thus, cluster-set configuration may allow high-volume training with high mechanical performance, which are factors that have relevant roles in the adaptation to resistance training. In this regard, volume plays a key role in muscle-endurance adaptation and in increase of strength.^{17,18,24} Similarly, the kinematics and kinetics (eg, velocity, work) associated with resistance exercise have been suggested to be relevant stimuli for the increase in strength and power or for induction of morphological adaptations.¹

Our findings show high intersubject variability of the MNR during CT, despite individualization of the resting time between repetitions, and are in line with those of a previous study.¹⁹ Neither recovery time between repetitions ($r = -.186$, $P = 0.632$) nor number of repetitions performed in TT ($r = .141$, $P = .718$) was correlated with the MNR for the CT. This suggests that muscle endurance in TT and muscle endurance in CT are sustained by different factors, so individualizing the rest pause did not reduce intersubject variability. Variability of the training response could be influenced by the type of exercise performed. Performance of multijoint resistance exercises such as squats requires relative effort in several muscle groups (synergists and stabilizing muscles) that vary with depth and load,²⁵ and thus the causes of failure could be different between subjects. In this regard, MNR in CT was only correlated with standard deviation of MPV in CT, suggesting that the capacity to perform the exercise in a wider range of velocities determines the intensity of muscle endurance in a cluster-set configuration.

Another interesting finding of this study was that while in TT 1RM and 4RM were negatively correlated with performed number of repetitions, this was not the case in CT. Negative correlation between 1RM and number of repetitions at a submaximal load has been previously reported,^{19,26–28} indicating that absolute load can exert a negative influence on muscle endurance due to higher occlusion of blood flow, greater accumulation of metabolites, or physiological adaptations that lead to improved maximum force.^{27–29}

As volume is a key factor in strength and high-intensity-exercise endurance development,^{16–18,24} set configuration employed in CT could be a useful strategy to carry out high-volume sessions of high intensity in athletes who have high levels of resistance performance.

There is a limitation of this study that needs to be addressed. The sessions were not randomized since TT was used to calculate resting time between repetitions for CT. Therefore, familiarization could have influenced the results. Nevertheless, there are several aspects that can rule out possible familiarization influences: Participants were well-trained judoists who habitually employed these exercises in their training regimens, mean concentric displacement was similar between sessions (Table 2), and finally there were no significant differences for MPV between best repetition of TT and first repetition in CT (0.43 $\pm$ 0.06 and 0.45 $\pm$ 0.08 m/s for TT and CT, respectively; $P = .181$), suggesting that the starting condition for every training session was similar.

Practical Applications

The current study showed that CT is useful to improve volume and to ensure sustainability of mechanical stimuli when resistance exercises are performed with high loads. Coaches should consider this type of set configuration for developing high-intensity muscle endurance. This is an objective of interest for specialties such as weightlifting, wrestling, or judo.

Conclusions

This study demonstrates that cluster set is a useful configuration of resistance exercise to achieve workouts of high volumes at high intensity. Second, individualizing pauses between repetitions with reference to muscle endurance assessed in TT did not reduce variability in MNR of CT, and thus further studies are necessary to identify mechanisms or subjects profiles that influence this variability. Finally, unlike TT, MNR of CT was not affected by absolute load. Thus, cluster-set configuration is useful for high-volume performance of high-intensity exercises.

References

1. Crewther B, Cronin J, Keogh J. Possible stimuli for strength and power adaptation: acute mechanical responses. *Sports Med.* 2005;35(11):967–989. PubMed doi:10.2165/00007256-200535110-00004

2. Denton J, Cronin JB. Kinematic, kinetic, and blood lactate profiles of continuous and intraset rest loading schemes. *J Strength Cond Res.* 2006;20(3):528–534. PubMed
3. Tran QT, Docherty D, Behm D. The effects of varying time under tension and volume load on acute neuromuscular responses. *Eur J Appl Physiol.* 2006;98(4):402–410. PubMed doi:10.1007/s00421-006-0297-3
4. Sanchez-Medina L, Gonzalez-Badillo JJ. Velocity loss as an indicator of neuromuscular fatigue during resistance training. *Med Sci Sports Exerc.* 2011;43(9):1725–1734. PubMed doi:10.1249/MSS.0b013e318213f880
5. Willardson JM, Burkett LN. The effect of rest interval length on the sustainability of squat and bench press repetitions. *J Strength Cond Res.* 2006;20(2):400–403. PubMed
6. Izquierdo M, Gonzalez-Badillo JJ, Hakkinen K, et al. Effect of loading on unintentional lifting velocity declines during single sets of repetitions to failure during upper and lower extremity muscle actions. *Int J Sports Med.* 2006;27(9):718–724. PubMed doi:10.1055/s-2005-872825
7. Willardson JM, Simao R, Fontana FE. The effect of load reductions on repetition performance for commonly performed multijoint resistance exercises. *J Strength Cond Res.* 2012;26(11):2939–2945. PubMed doi:10.1519/JSC.0b013e3182430170
8. Ratamess NA, Falvo MJ, Mangine GT, Hoffman JR, Faigenbaum AD, Kang J. The effect of rest interval length on metabolic responses to the bench press exercise. *Eur J Appl Physiol.* 2007;100(1):1–17. PubMed doi:10.1007/s00421-007-0394-y
9. Haff G, Hobbs R, Haff E, Sands W, Pierce K, Stone M. Cluster training: a novel method for introducing training program variation. *Strength Cond J.* 2008;30(1):67–76. doi:10.1519/SSC.0b013e31816383e1
10. Haff GG, Whitley A, McCoy LB, et al. Effects of different set configurations on barbell velocity and displacement during a clean pull. *J Strength Cond Res.* 2003;17(1):95–103. PubMed
11. Hansen KT, Cronin JB, Pickering SL, Newton MJ. Does cluster loading enhance lower body power development in preseason preparation of elite rugby union players? *J Strength Cond Res.* 2011;25(8):2118–2126. PubMed doi:10.1519/JSC.0b013e318220b6a3
12. Hansen KT, Cronin JB, Newton MJ. The effect of cluster loading on force, velocity, and power during ballistic jump squat training. *Int J Sports Physiol Perform.* 2011;6(4):455–468. PubMed
13. Hardee JP, Triplett NT, Utter AC, Zwetsloot KA, McBride JM. Effect of interrepetition rest on power output in the power clean. *J Strength Cond Res.* 2012;26(4):883–889. PubMed doi:10.1519/JSC.0b013e3182474370
14. Iglesias-Soler E, Carballeira E, Sanchez-Otero T, Mayo X, Jimenez A, Chapman ML. Acute effects of distribution of rest between repetitions. *Int J Sports Med.* 2012;33(5):351–358. PubMed doi:10.1055/s-0031-1299699
15. Hardee JP, Lawrence MM, Zwetsloot KA, Triplett NT, Utter AC, McBride JM. Effect of cluster set configurations on power clean technique. *J Sports Sci.* 2013;31(5):488–496. PubMed doi:10.1080/02640414.2012.736633
16. Gonzalez-Badillo JJ, Izquierdo M, Gorostiaga EM. Moderate volume of high relative training intensity produces greater strength gains compared with low and high volumes in competitive weightlifters. *J Strength Cond Res.* 2006;20(1):73–81. PubMed
17. Stone M, Stone M, Sands W, et al. Maximum strength and strength training—a relationship to endurance? *Strength Cond J.* 2006;28(3):44–53.
18. Munn J, Herbert RD, Hancock MJ, Gandevia SC. Resistance training for strength: effect of number of sets and contraction speed. *Med Sci Sports Exerc.* 2005;37(9):1622–1626. PubMed doi:10.1249/01.mss.0000177583.41245.f8
19. Iglesias E, Boulosa DA, Dopico X, Carballeira E. Analysis of factors that influence the maximum number of repetitions in two upper-body resistance exercises: curl biceps and bench press. *J Strength Cond Res.* 2010;24(6):1566–1572. PubMed doi:10.1519/JSC.0b013e3181d8eabe
20. Lawton TW, Cronin JB, Lindsell RP. Effect of interrepetition rest intervals on weight training repetition power output. *J Strength Cond Res.* 2006;20(1):172–176. PubMed
21. Bosco C, Belli A, Astrua M, et al. A dynamometer for evaluation of dynamic muscle work. *Eur J Appl Physiol Occup Physiol.* 1995;70(5):379–386.
22. Nakagawa S, Cuthill IC. Effect size, confidence interval and statistical significance: a practical guide for biologists. *Biol Rev Camb Philos Soc.* 2007;82(4):591–605. PubMed doi:10.1111/j.1469-185X.2007.00027.x
23. Lawton T, Cronin J, Drinkwater E, Lindsell R, Pyne D. The effect of continuous repetition training and intra-set rest training on bench press strength and power. *J Sports Med Phys Fitness.* 2004;44(4):361–367. PubMed
24. Willardson JM, Emmett J, Oliver JA, Bressel E. Effect of short-term failure versus nonfailure training on lower body muscular endurance. *Int J Sports Physiol Perform.* 2008;3(3):279–293. PubMed
25. Bryant MA, Kennedy MD, Carey JP, Chiu LZ. Effect of squat depth and barbell load on relative muscular effort in squatting. *J Strength Cond Res.* 2012;26(10):2820–2828. PubMed doi:10.1519/JSC.0b013e31826791a7
26. Faigenbaum AD, Ratamess NA, McFarland J, et al. Effect of rest interval length on bench press performance in boys, teens, and men. *Pediatr Exerc Sci.* 2008;20(4):457–469. PubMed
27. Ratamess NA, Chiarello CM, Sacco AJ, et al. The effects of rest interval length on acute bench press performance: the influence of gender and muscle strength. *J Strength Cond Res.* 2012;26(7):1817–1826. PubMed doi:10.1519/JSC.0b013e31825bb492
28. Panissa V, Azevedo Neto R, Julio U, et al. Maximum number of repetitions, total weight lifted and neuromuscular fatigue in individuals with different training backgrounds. *Biol Sport.* 2013;30(2):131–136. doi:10.5604/20831862.1044458
29. Izquierdo M, Ibanez J, Calbet JA, et al. Neuromuscular fatigue after resistance training. *Int J Sports Med.* 2009;30(8):614–623. PubMed doi:10.1055/s-0029-1214379

Copyright of International Journal of Sports Physiology & Performance is the property of Human Kinetics Publishers, Inc. and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.